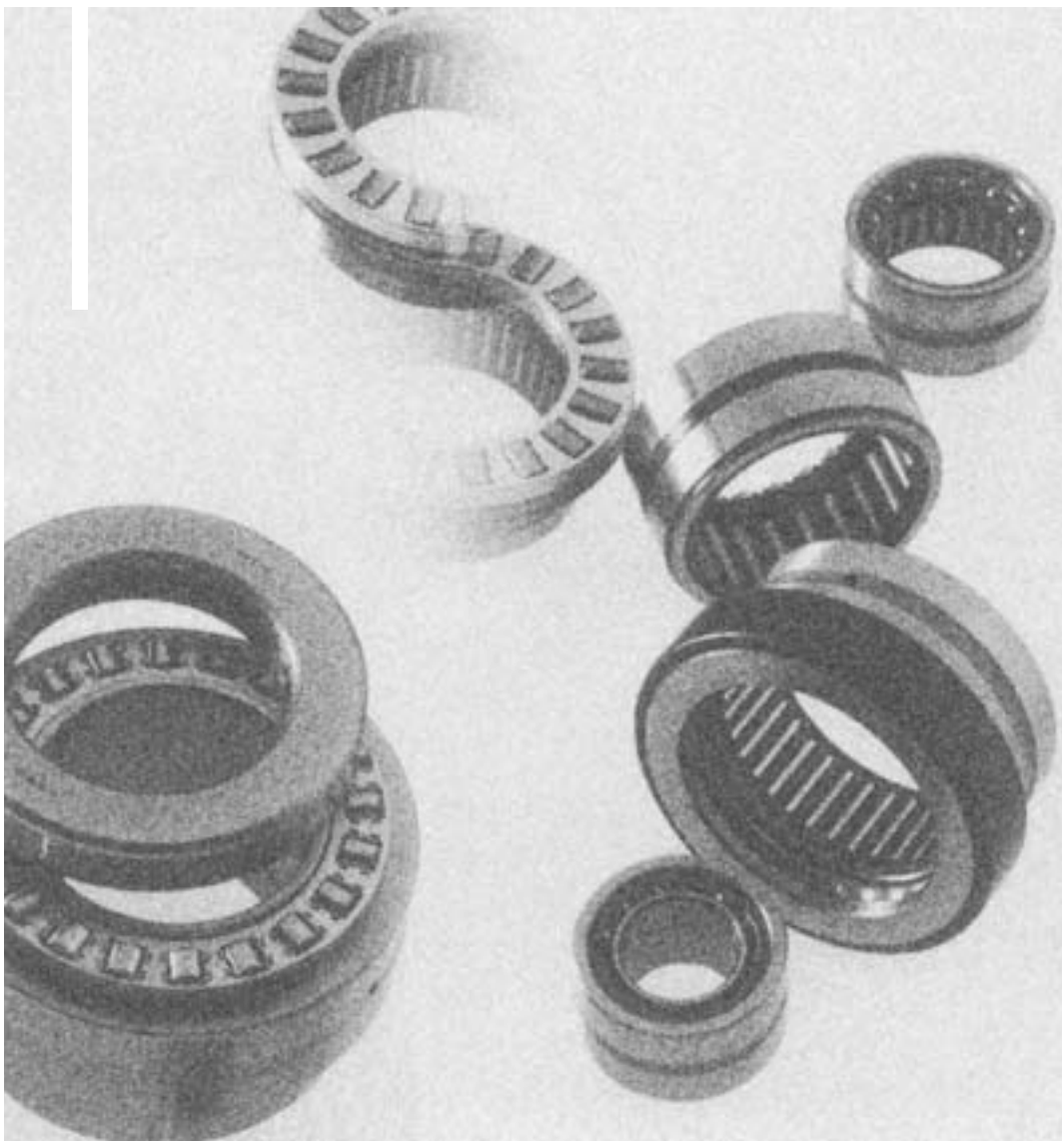
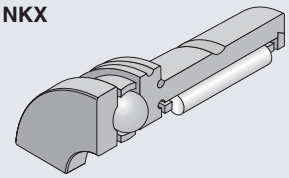
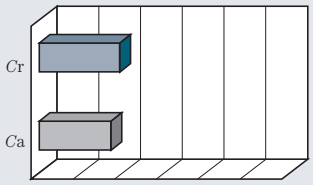
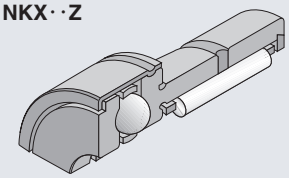
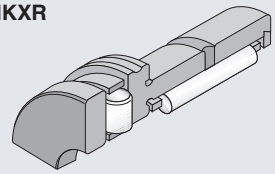
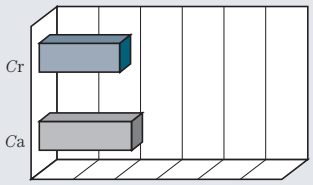
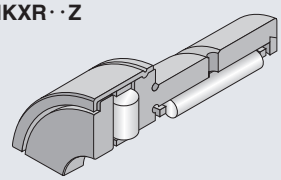
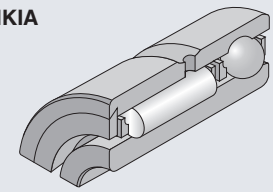
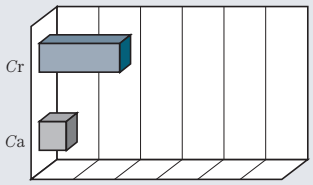
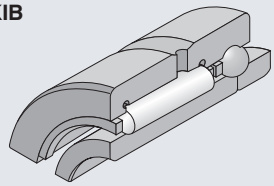
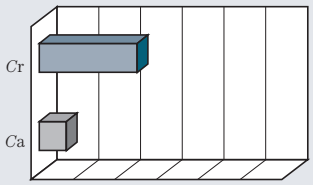
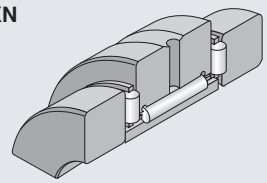
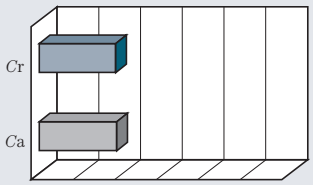
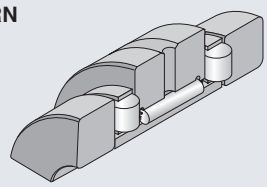
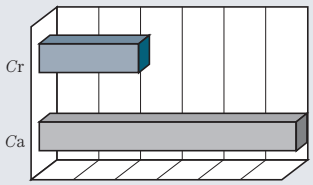
Complex Bearings



Complex Bearings

NTN complex bearing is composed of a radial needle roller bearing to support radial load and a thrust ball bearing or a thrust roller bearing to support axial load which are integrated into one bearing unit. Comparing with individual use of a radial bearing and a thrust bearing for the same purpose, this complex bearing saves the required installation space and thereby can contribute to making the overall construction of an equipment more compact.

Bearing type	Applied shaft diameter (mm)	Load capacity	Composition of nominal bearing number
Type NKX 	ϕ10～ϕ70		**NKX 20 T2** Tail code T2: Resin cage Dimension code (shaft diameter) Type code
Type NKX··Z 			**NKX 20 T2 Z** Tail code T2: Resin cage Z: with dust-proof cover Dimension code (shaft diameter) Type code
Type NKXR 	ϕ15～ϕ50		**NKXR 20 T2** Tail code T2: Resin cage Dimension code (shaft diameter) Type code
Type NKXR··Z 			**NKXR 20 T2Z** Tail code T2: Resin cage Z: with dust-proof cover Dimension code (shaft diameter) Type code
Type NKIA 	ϕ15～ϕ70		**NKIA 59 04** Bore diameter code Dimension series code Type code
Type NKIB 	ϕ15～ϕ70		**NKIB 59 04 R** Tail code R: outer ring with rib Bore diameter code Dimension series code Type code
Type AXN 	ϕ20～ϕ50		**AXN 20 52** Outer diameter code Bore diameter code Type code
Type ARN 	ϕ20～ϕ70		**ARN 20 62** Outer diameter code Bore diameter code Type code

※Each listed load capacity is subject to reference bearing bore diameter of ϕ20.

Bearing components	Handling characteristic	Features
Inscribed circle diameter (shaft dia.) : ϕ20 Radial bearing : Needle roller type Iron cage Thrust bearing : Ball type Resin cage Dust-proof cover : without Thrust plate : Separable	△ Separable thrust plate Without dust-proof cover	Can be applied to high speed running under light axial load, due to use of ball type thrust bearing.
Inscribed circle diameter (shaft dia.) : ϕ20 Radial bearing : Needle roller type Iron cage Thrust bearing : Ball type Resin cage Dust-proof cover : with Thrust plate : Non-separable, integral type	○ Integral thrust plate With dust-proof cover	This type is identical to NKZ, but its thrust bearing is provided with dust-proof cover to prevent grease scattering. Furthermore, the handling characteristic is good because the dust-proof cover makes the thrust plate non-separable integral.
Inscribed circle diameter (shaft dia.) : ϕ20 Radial bearing : Needle roller type Iron cage Thrust bearing : Ball type Resin cage Dust-proof cover : without Thrust plate : Separable	△ Separable thrust plate Without dust-proof cover	Due to the use of needle roller bearing as its thrust bearing, this type can support greater axial load than NKX.
Inscribed circle diameter (shaft dia.) : ϕ20 Radial bearing : Needle roller type Iron cage Thrust bearing : Ball type Resin cage Dust-proof cover : with Thrust plate : Non-separable, integral type	○ Integral thrust plate With dust-proof cover	This type is identical to NKXR, but its thrust bearing is provided with dust-proof cover to prevent grease scattering. Furthermore, the handling characteristic is good because the dust-proof cover makes the thrust plate non-separable and integral.
Inscribed circle diameter (shaft dia.) : ϕ20 Radial bearing : Needle roller type Thrust bearing : Angular type	◎ Integral angular bearing Inner ring separable	This type is composed of needle roller bearing and angular bearing which were integrated into one unit. The angular bearing supports axial load. Hence, this type can support one-way axial load only. Best-suited to high speed and light load application.
Inscribed circle diameter (shaft dia.) : ϕ20 Radial bearing : Needle roller type Outer ring with rib Thrust bearing : Three-point contact angular type	○ Double inner ring type Inner ring separable	This type is composed of needle roller bearing and three-point contact angular bearing which were integrated into one unit. Unlike KN1A, this type can support bi-directional axial load.
Inscribed circle diameter (shaft dia.) : ϕ20 Outer ring outer diameter : 52 Radial bearing : Needle roller type Thrust bearing : Needle roller type	△ Separable	This type is a special-purposed bearing to support a precision ball screw. This type uses the side face of the radial bearing as the raceway surface of thrust bearing. This can support axial load from both directions.
Inscribed circle diameter (shaft dia.) : ϕ20 Outer ring outer diameter : 62 Radial bearing : Needle roller type Thrust bearing : Cylindrical roller type	△ Separable	This type is a special-purposed bearing to support a precision ball screw. This can support bi-directional axial load. This type is identical to AXN, but its axial load capacity is greater.

Bearing Accuracy

Regarding the dimensional tolerances for radial needle roller bearing, refer to **Table 4.3** of 4. "**Bearing Tolerances**" (page A-26) for others than the characteristics described in Dimensions Table and to **Table 4.4** (page A-26) for thrust bearing tolerances respectively. Manufacture of bearing **Type AXN** and **ARN** is limited to those of JIS accuracy class-5 and -4 only. (Allowable radial run-out values for Type AXN and ARN is as specified in **Table 1**.)

Table 1 Allowable radial run-out values for Type AXN and ARN

Unit: μm

Nominal bearing bore dia. d or nominal bearing outer dia. D (mm)		Radial runout of ❶ radial inner ring K_{ia}		Allowable radial ❷ run-out for outer ring K_{ea}	
Over	Incl.	Class 5	Class 4	Class 5	Class 4
18	30	4	3	—	—
30	50	5	4	—	—
50	80	5	4	8	5
80	120	—	—	10	6
120	150	—	—	11	7

❶ To be determined based on "d".
❷ To be determined based on "D".

Radial clearance

The respective radial clearances of **Type NKX+IR**, **NKXR+IR**, **NKIA** and **NKIB** shall comply with the interchangeable clearances specified in **Table 5.1** (1) of 5.1 "**Radial Clearances of Bearings**" (page A-30). The standard bearings are manufactured subject to the regular clearances.

The radial clearances of **Type AXN** and **ARN** is special clearance, as specified in applicable Dimensions Table.

Bearing fits

Radial needle roller bearings shall be fitted on shaft/in housing in compliance with **Table 2**. The thrust bearing washers of **Type NKX** and **NKXR** shall be fitted in a housing with hole diameter larger by 0.5mm or over than the washer outer diameter D_1 or D_2. For **Type NKIA** and **NKIB** it is not allowed to make the interference greater than k5/M6.

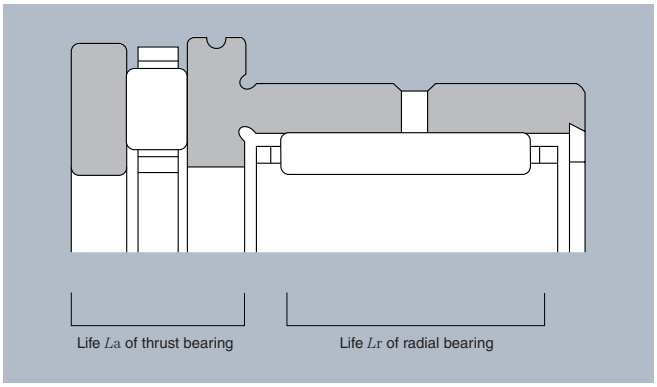
Table 2 Bearing fit on shaft and in housing

Bearing type	Shaft	Housing
Series NKX, NKXR	k5	K6 (M6)
Series NKIA, NKIB	k5	M6
Series AXN, ARN	j5	J6

Remarks: The parenthesized housing code is applied, where high rigidity is required.

Calculation of rated life

The rated life (L) of any complex bearing is determined as follows. Determine individually the rated life (L_r) of radial needle roller bearing from radial load acting thereon and the rated life (L_a) of thrust bearing from axial load acting thereon. And determine the rated life (L) from these determined rated lives (L_r, L_a) using the following formula

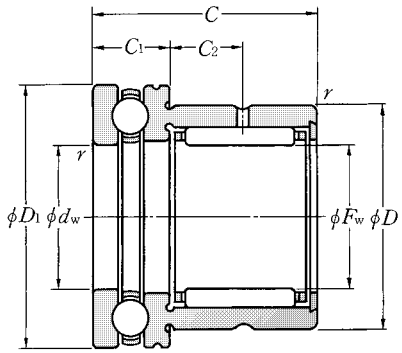


Rated life L of complex bearing

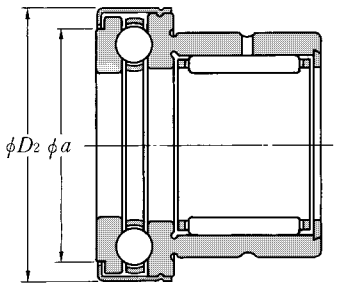
$$L=\frac{1}{\left(\frac{1}{L_r^{1.1}}+\frac{1}{L_a^{1.1}}\right)^{0.91}} \quad \cdots\cdots(1)$$

Without inner ring

Type NKX
Type NKX··Z◦



Type NKX
(Open type)

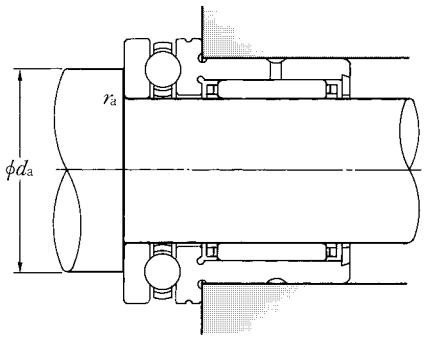


Type NKX··Z
(With cover)

F_w 10～70mm

Boundary dimensions mm											Basic load ratings N		Basic load ratings kgf	
											dynamic	static	dynamic	static
											Radial			
F_w	d_w	D	D_1	D_2	C 0 −0.25	C_1 0 −0.20	C_2	a	$r_{s\,min}$ 1)	C_r	C_{or}	C_r	C_{or}
10 +0.022 +0.013	10 +0.040 +0.025	19	24	25	23	9	6.5	19.7	0.3	5 450	6 450	555	660
12 +0.027 +0.016	12 +0.050 +0.032	21	26	27	23	9	6.5	21.7	0.3	6 000	7 700	615	785
15 +0.027 +0.016	15 +0.050 +0.032	24	28	29	23	9	6.5	23.7	0.3	8 250	10 200	840	1 040
17 +0.027 +0.016	17 +0.050 +0.032	26	30	31	25	9	8	25.7	0.3	10 400	14 400	1 060	1 460
20 +0.033 +0.020	20 +0.061 +0.040	30	35	36	30	10	10.5	30.7	0.3	16 400	27 100	1 670	2 760
25 +0.033 +0.020	25 +0.061 +0.040	37	42	43	30	11	9.5	37.7	0.6	14 200	24 000	1 450	2 450
30 +0.033 +0.020	30 +0.061 +0.040	42	47	48	30	11	9.5	42.7	0.6	22 300	39 500	2 280	4 000
35 +0.041 +0.025	35 +0.075 +0.050	47	52	53	30	12	9	47.7	0.6	20 000	36 000	2 040	3 650
40 +0.041 +0.025	40 +0.075 +0.050	52	60	61	32	13	10	55.7	0.6	25 900	52 500	2 650	5 350
45 +0.041 +0.025	45 +0.075 +0.050	58	65	66.5	32	14	9	60.5	0.6	27 600	59 000	2 810	6 000
50 +0.041 +0.025	50 +0.075 +0.050	62	70	71.5	35	14	10	65.5	0.6	27 900	62 000	2 850	6 300
60 +0.049 +0.030	60 +0.090 +0.060	72	85	86.5	40	17	12	80.5	1	29 800	71 500	3 050	7 300
70 +0.049 +0.030	70 +0.090 +0.060	85	95	96.5	40	18	11	90.5	1	36 500	86 000	3 700	8 750

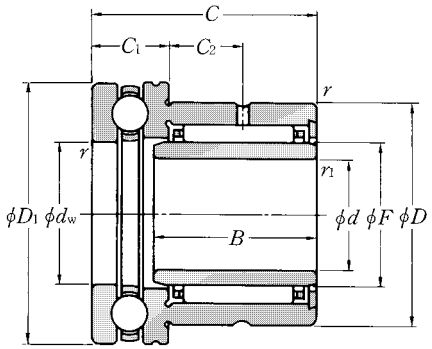
Note 1) Allowable minimum chamfer dimension r.



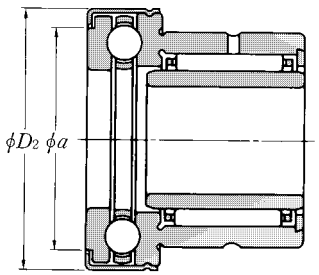
Basic load ratings dynamic N Axial C_a	Basic load ratings static N Axial C_{oa}	Basic load ratings dynamic kgf Axial C_a	Basic load ratings static kgf Axial C_{oa}	Limiting dimensions r/min grease	Limiting dimensions r/min oil	Bearing numbers Type NKX	Bearing numbers Type NKX··Z	Abutment dimensions mm d_a min	Abutment dimensions mm r_{as} max	Mass (approx.) kg Type NKX	Mass (approx.) kg Type NKX··Z
10 000	14 000	1 020	1 420	6 700	9 500	**NKX10T2**	**NKX10T2Z**	18	0.3	0.037	0.039
10 300	15 400	1 050	1 570	6 400	9 200	**NKX12T2**	**NKX12T2Z**	20	0.3	0.042	0.044
10 500	16 800	1 070	1 710	6 200	8 800	**NKX15T2**	**NKX15T2Z**	23	0.3	0.044	0.048
10 800	18 200	1 100	1 850	6 000	8 500	**NKX17T2**	**NKX17T2Z**	25	0.3	0.051	0.056
14 200	24 700	1 450	2 520	5 200	7 500	**NKX20T2**	**NKX20T2Z**	29	0.3	0.085	0.090
19 600	37 000	1 990	3 800	4 600	6 500	**NKX25T2**	**NKX25T2Z**	35	0.6	0.125	0.132
20 400	42 000	2 080	4 300	4 300	6 200	**NKX30T2**	**NKX30T2Z**	40	0.6	0.140	0.148
20 400	44 500	2 080	4 550	3 900	5 600	**NKX35T2**	**NKX35T2Z**	45	0.6	0.167	0.175
26 900	63 000	2 740	6 400	3 500	5 000	**NKX40**	**NKX40Z**	52	0.6	0.216	0.225
27 900	69 000	2 840	7 050	3 200	4 600	**NKX45**	**NKX45Z**	57	0.6	0.252	0.265
28 800	75 500	2 930	7 700	3 100	4 500	**NKX50**	**NKX50Z**	62	0.6	0.302	0.318
41 500	113 000	4 200	11 500	2 600	3 700	**NKX60**	**NKX60Z**	75	1	0.465	0.484
43 000	127 000	4 400	12 900	2 400	3 400	**NKX70**	**NKX70Z**	85	1	0.612	0.635

With inner ring

Type NKX＋IR
Type NKX··Z＋IR



Type NKX＋IR
(Open type)

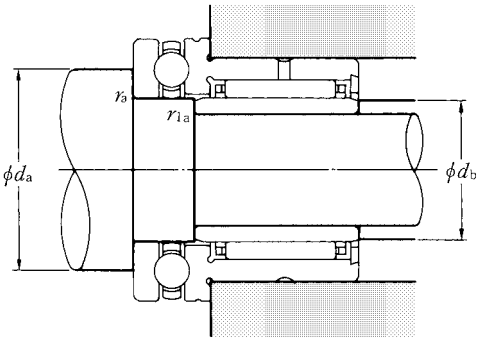


Type NKX··Z＋IR
(With cover)

d 7～60mm

Boundary dimensions												Basic load ratings							
												dynamic	static	dynamic	static	dynamic	static	dynamic	static
												N		kgf		N		kgf	
												Radial				Axial			
d	*dw*	*D*	*D1*	*D2*	*B*	*C* 0 −0.25	*C1* 0 −0.20	*C2*	*F*	*a*	*r*s min 1)	*r*1s min 1)	*Cr*	*Cor*	*Cr*	*Cor*	*Ca*	*Coa*	*Ca*	*Coa*
7	10 +0.040 +0.025	19	24	25	16	23	9	6.5	10	19.7	0.3	0.3	5 450	6 450	555	660	10 000	14 000	1 020	1 420
9	12 +0.050 +0.032	21	26	27	16	23	9	6.5	12	21.7	0.3	0.3	6 000	7 700	615	785	10 300	15 400	1 050	1 570
12	15 +0.050 +0.032	24	28	29	16	23	9	6.5	15	23.7	0.3	0.3	8 250	10 200	840	1 040	10 500	16 800	1 070	1 710
14	17 +0.050 +0.032	26	30	31	17	25	9	8	17	25.7	0.3	0.3	10 400	14 400	1 060	1 460	10 800	18 200	1 100	1 850
17	20 +0.061 +0.040	30	35	36	20	30	10	10.5	20	30.7	0.3	0.3	16 400	27 100	1 670	2 760	14 200	24 700	1 450	2 520
20	25 +0.061 +0.040	37	42	43	20	30	11	9.5	25	37.7	0.6	0.3	14 200	24 000	1 450	2 450	19 600	37 000	1 990	3 800
25	30 +0.061 +0.040	42	47	48	20	30	11	9.5	30	42.7	0.6	0.3	22 300	39 500	2 280	4 000	20 400	42 000	2 080	4 300
30	35 +0.075 +0.050	47	52	53	20	30	12	9	35	47.7	0.6	0.3	20 000	36 000	2 040	3 650	20 400	44 500	2 080	4 550
35	40 +0.075 +0.050	52	60	61	20	32	13	10	40	55.7	0.6	0.3	25 900	52 500	2 650	5 350	26 900	63 000	2 740	6 400
40	45 +0.075 +0.050	58	65	66.5	20	32	14	9	45	60.5	0.6	0.3	27 600	59 000	2 810	6 000	27 900	69 000	2 840	7 050
45	50 +0.075 +0.050	62	70	71.5	25	35	14	10	50	65.5	0.6	0.6	27 900	62 000	2 850	6 300	28 800	75 500	2 930	7 700
50	60 +0.090 +0.060	72	85	86.5	25	40	17	12	60	80.5	1	1	29 800	71 500	3 050	7 300	41 500	113 000	4 200	11 500
60	70 +0.090 +0.060	85	95	96.5	25	40	18	11	70	90.5	1	1	36 500	86 000	3 700	8 750	43 000	127 000	4 400	12 900

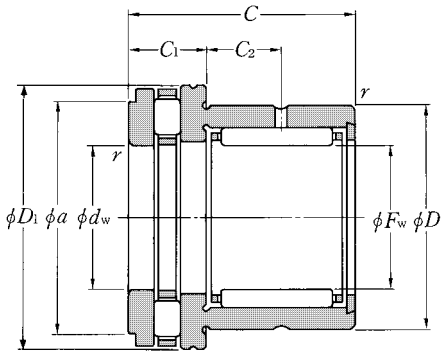
Note 1) Allowable minimum chamfer dimension *r*.
Remarks: Nominal code number of inner ring comprises IR. Bore diameter×outer diameter×width.



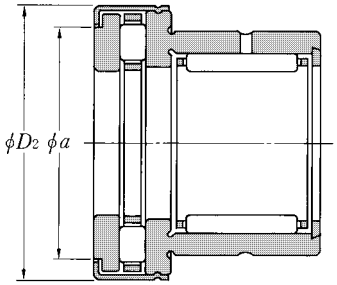
Limiting dimensions r/min grease	oil	Bearing numbers Type NKX+IR	Type NKX··Z+IR	Abutment dimensions mm d_a min	d_b min	r_{as} max	r_{1as} max	Mass (approx.) kg Type NKX+IR	Type NKX··Z+IR
6 700	9 500	NKX10T2+IR 7×10×16	NKX10T2Z+IR 7×10×16	18	9	0.3	0.3	0.042	0.044
6 400	9 200	NKX12T2+IR 9×12×16	NKX12T2Z+IR 9×12×16	20	11	0.3	0.3	0.048	0.050
6 200	8 800	NKX15T2+IR12×15×16	NKX15T2Z+IR12×15×16	23	14	0.3	0.3	0.052	0.056
6 000	8 500	NKX17T2+IR14×17×17	NKX17T2Z+IR14×17×17	25	16	0.3	0.3	0.061	0.066
5 200	7 500	NKX20T2+IR17×20×20	NKX20T2Z+IR17×20×20	29	19	0.3	0.3	0.099	0.104
4 600	6 500	NKX25T2+IR20×25×20	NKX25T2Z+IR20×25×20	35	22	0.6	0.3	0.152	0.159
4 300	6 200	NKX30T2+IR25×30×20	NKX30T2Z+IR25×30×20	40	27	0.6	0.3	0.173	0.181
3 900	5 600	NKX35T2+IR30×35×20	NKX35T2Z+IR30×35×20	45	32	0.6	0.3	0.205	0.213
3 500	5 000	NKX40 +IR35×40×20	NKX40Z +IR35×40×20	52	37	0.6	0.3	0.260	0.269
3 200	4 600	NKX45 +IR40×45×20	NKX45Z +IR40×45×20	57	42	0.6	0.3	0.293	0.316
3 100	4 500	NKX50 +IR45×50×25	NKX50Z +IR45×50×25	62	48	0.6	0.6	0.373	0.389
2 600	3 700	NKX60 +IR50×60×25	NKX60Z +IR50×60×25	75	55	1	1	0.635	0.654
2 400	3 400	NKX70 +IR60×70×25	NKX70Z +IR60×70×25	85	65	1	1	0.814	0.837

Without inner ring

Type NKXR
Type NKXR··Z◦



Type NKXR
(Open type)

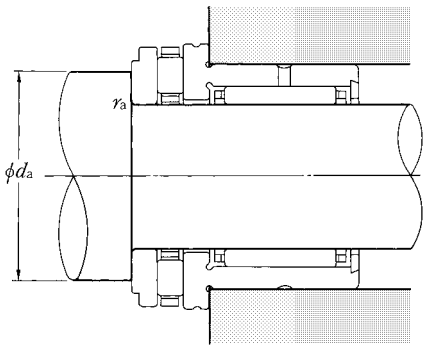


Type NKXR··Z
(With cover)

F_w 15～50mm

Boundary dimensions mm											Basic load ratings N		dynamic kgf	static kgf
F_w		d_w		D	D_1	D_2	C 0 −0.25	C_1 0 −0.20	C_2	a	$r_{s\,min}$ 1)	Radial dynamic C_r	Radial static C_{or}	Radial dynamic C_r	Radial static C_{or}
15	+0.027 +0.016	15	+0.050 +0.032	24	28	29	23	9	6.5	23.7	0.3	8 250	10 200	840	1 040
17	+0.027 +0.016	17	+0.050 +0.032	26	30	31	25	9	8	25.7	0.3	10 400	14 400	1 060	1 460
20	+0.033 +0.020	20	+0.061 +0.040	30	35	36	30	10	10.5	30.7	0.3	16 400	27 100	1 670	2 760
25	+0.033 +0.020	25	+0.061 +0.040	37	42	43	30	11	9.5	37.7	0.6	14 200	24 000	1 450	2 450
30	+0.033 +0.020	30	+0.061 +0.040	42	47	48	30	11	9.5	42.7	0.6	22 300	39 500	2 280	4 000
35	+0.041 +0.025	35	+0.075 +0.050	47	52	53	30	12	9	47.7	0.6	20 000	36 000	2 040	3 650
40	+0.041 +0.025	40	+0.075 +0.050	52	60	61	32	13	10	55.7	0.6	25 900	52 500	2 650	5 350
45	+0.041 +0.025	45	+0.075 +0.050	58	65	66.5	32	14	9	60.5	0.6	27 600	59 000	2 810	6 000
50	+0.041 +0.025	50	+0.075 +0.050	62	70	71.5	35	14	10	65.5	0.6	27 900	62 000	2 850	6 300

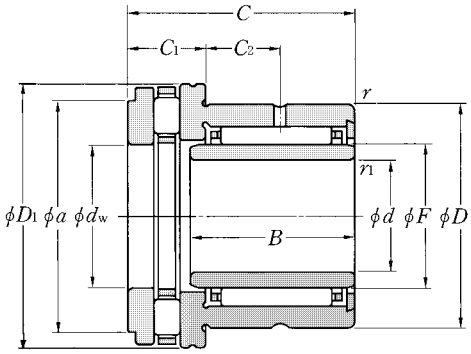
Note 1) Allowable minimum chamfer dimension r.



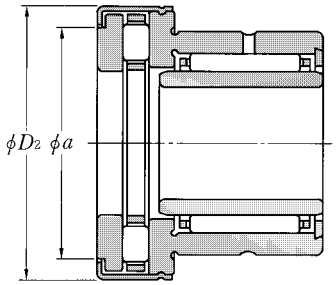
Basic load ratings N Axial dynamic C_a	Basic load ratings N Axial static C_{oa}	Basic load ratings kgf Axial dynamic C_a	Basic load ratings kgf Axial static C_{oa}	Limiting dimensions r/min grease	Limiting dimensions r/min oil	Bearing numbers Type NKXR	Bearing numbers Type NKXR··Z*	Abutment dimensions mm d_a min	Abutment dimensions mm r_{as} max	Mass (approx.) kg Type NKXR	Mass (approx.) kg Type NKXR··Z
12 900	28 600	1 310	2 920	2 800	11 000	**NKXR15T2**	**NKXR15T2Z**	25	0.3	0.048	0.052
13 400	31 000	1 370	3 150	2 500	10 000	**NKXR17T2**	**NKXR17T2Z**	27	0.3	0.050	0.053
20 200	46 500	2 060	4 700	2 100	8 500	**NKXR20T2**	**NKXR20T2Z**	32	0.3	0.090	0.095
27 300	68 000	2 780	6 900	1 800	7 000	**NKXR25T2**	**NKXR25T2Z**	39	0.6	0.128	0.135
27 800	72 500	2 840	7 400	1 500	6 000	**NKXR30T2**	**NKXR30T2Z**	44	0.6	0.162	0.169
31 000	87 000	3 150	8 900	1 400	5 500	**NKXR35T2**	**NKXR35T2Z**	49	0.6	0.184	0.195
43 000	121 000	4 350	12 400	1 200	4 800	**NKXR40T2**	**NKXR40T2Z**	56	0.6	0.226	0.237
45 500	135 000	4 650	13 800	1 100	4 400	**NKXR45T2**	**NKXR45T2Z**	61	0.6	0.267	0.286
48 000	150 000	4 900	15 300	1 000	4 000	**NKXR50T2**	**NKXR50T2Z**	66	0.6	0.309	0.329

With inner ring

Type NKXR＋IR
Type NKXR··Z＋IR



Type NKXR＋IR
(Open type)

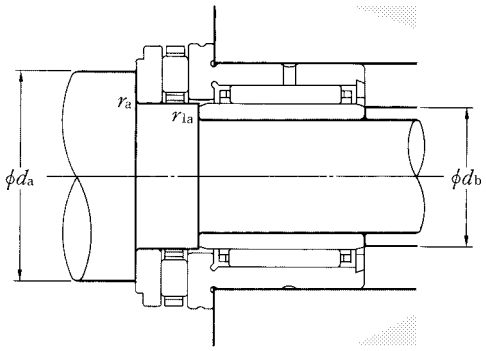


Type NKXR··Z＋IR
(With cover)

d 12～45mm

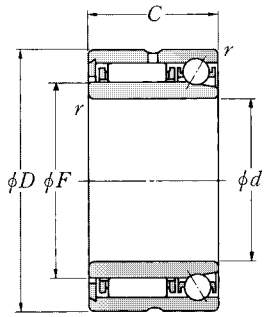
Boundary dimensions mm													Basic load ratings dynamic N	static N	dynamic kgf	static kgf	dynamic N	static N	dynamic kgf	static kgf
													Radial				Axial			
d	d_w	D	D_1	D_2	B	C $^{0}_{-0.25}$	C_1 $^{0}_{-0.20}$	C_2	F	a	$r_{s\,min}$ 1)	$r_{1s\,min}$ 1)	C_r	C_{or}	C_r	C_{or}	C_a	C_{oa}	C_a	C_{oa}
12	15 $^{+0.050}_{+0.032}$	24	28	29	16	23	9	6.5	15	23.7	0.3	0.3	8 250	10 200	840	1 040	12 900	28 600	1 310	2 920
14	17 $^{+0.050}_{+0.032}$	26	30	31	17	25	9	8	17	25.7	0.3	0.3	10 400	14 400	1 060	1 460	13 400	31 000	1 370	3 150
17	20 $^{+0.061}_{+0.040}$	30	35	36	20	30	10	10.5	20	30.7	0.3	0.3	16 400	27 100	1 670	2 760	20 200	46 500	2 060	4 700
20	25 $^{+0.061}_{+0.040}$	37	42	43	20	30	11	9.5	25	37.7	0.6	0.3	14 200	24 000	1 450	2 450	27 300	68 000	2 780	6 900
25	30 $^{+0.061}_{+0.040}$	42	47	48	20	30	11	9.5	30	42.7	0.6	0.3	22 300	39 500	2 280	4 000	27 800	72 500	2 840	7 400
30	35 $^{+0.075}_{+0.050}$	47	52	53	20	30	12	9	35	47.7	0.6	0.3	20 000	36 000	2 040	3 650	31 000	87 000	3 150	8 900
35	40 $^{+0.075}_{+0.050}$	52	60	61	20	32	13	10	40	55.7	0.6	0.3	25 900	52 500	2 650	5 350	43 000	121 000	4 350	12 400
40	45 $^{+0.075}_{+0.050}$	58	65	66.5	20	32	14	9	45	60.5	0.6	0.3	27 600	59 000	2 810	6 000	45 500	135 000	4 650	13 800
45	50 $^{+0.075}_{+0.050}$	62	70	71.5	25	35	14	10	50	65.5	0.6	0.6	27 900	62 000	2 850	6 300	48 000	150 000	4 900	15 300

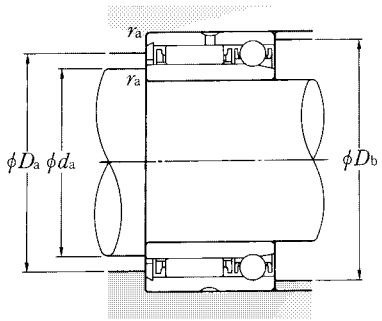
Note 1) Allowable minimum chamfer dimension r or r_1.
Remarks: Nominal code number of inner ring comprises IR. Bore diameter×outer diameter×width.



Limiting dimensions r/min grease	oil	Bearing numbers Type NKXR+IR	Type NKXR··Z+IR	Abutment dimensions mm d_a min	d_b min	r_{as} max	r_{1as} max	Mass (approx.) kg Type NKXR+IR	Type NKXR··Z+IR
2 800	11 000	NKXR15T2+IR12×15×16	NKXR15T2Z+IR12×15×16	25	14	0.3	0.3	0.056	0.060
2 500	10 000	NKXR17T2+IR14×17×17	NKXR17T2Z+IR14×17×17	27	16	0.3	0.3	0.060	0.063
2 100	8 500	NKXR20T2+IR17×20×20	NKXR20T2Z+IR17×20×20	32	19	0.3	0.3	0.105	0.110
1 800	7 000	NKXR25T2+IR20×25×20	NKXR25T2Z+IR20×25×20	39	22	0.6	0.3	0.155	0.162
1 500	6 000	NKXR30T2+IR25×30×20	NKXR30T2Z+IR25×30×20	44	27	0.6	0.3	0.197	0.202
1 400	5 500	NKXR35T2+IR30×35×20	NKXR35T2Z+IR30×35×20	49	32	0.6	0.3	0.224	0.235
1 200	4 800	NKXR40T2+IR35×40×20	NKXR40T2Z+IR35×40×20	56	37	0.6	0.3	0.270	0.281
1 100	4 400	NKXR45T2+IR40×45×20	NKXR45T2Z+IR40×45×20	61	42	0.6	0.3	0.318	0.337
1 000	4 000	NKXR50T2+IR45×50×25	NKXR50T2Z+IR45×50×25	66	48	0.6	0.6	0.379	0.400

Type NKIA59





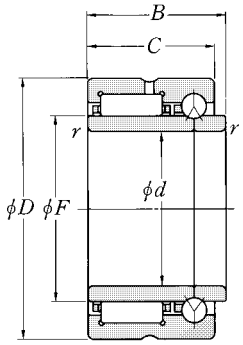
d 15～70mm

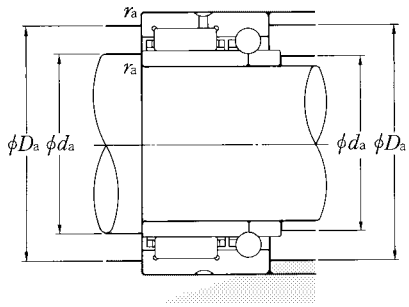
Boundary dimensions mm					Basic load ratings dynamic N	static N	dynamic kgf	static kgf	dynamic N	static N	dynamic kgf	static kgf	Limiting dimensions r/min	
					Radial				Axial				grease	oil
d	D	C	F	$r_{s\,min}$ 1)	C_r	C_{or}	C_r	C_{or}	C_a	C_{oa}	C_a	C_{oa}		
15	28	18	20	0.3	9 500	13 400	970	1 370	2 340	3 050	239	310	13 000	20 000
17	30	18	22	0.3	10 100	14 900	1 030	1 520	2 530	3 550	258	360	12 000	18 000
20	37	23	25	0.3	16 500	22 100	1 680	2 250	4 700	6 150	480	625	11 000	16 000
22	39	23	28	0.3	17 500	24 800	1 790	2 530	4 900	6 750	500	690	9 500	14 000
25	42	23	30	0.3	17 400	25 000	1 770	2 550	5 100	7 350	520	750	8 500	13 000
30	47	23	35	0.3	19 400	30 500	1 980	3 100	5 400	8 550	550	870	7 500	11 000
35	55	27	42	0.6	25 700	46 000	2 630	4 700	7 400	12 300	755	1 260	6 500	9 500
40	62	30	48	0.6	31 000	61 000	3 150	6 250	7 750	14 000	790	1 430	5 500	8 500
45	68	30	55	0.6	33 000	69 500	3 350	7 100	8 500	17 100	870	1 740	5 000	7 500
50	72	30	60	0.6	33 500	73 500	3 450	7 500	8 800	18 700	900	1 910	4 300	6 500
55	80	34	63	1	44 500	95 500	4 500	9 700	14 300	33 000	1 460	3 350	4 300	6 500
60	85	34	68	1	45 500	101 000	4 600	10 300	14 800	36 000	1 510	3 650	4 000	6 000
65	90	34	75	1	46 000	106 000	4 700	10 800	15 200	39 000	1 550	4 000	3 700	5 500
70	100	40	80	1	62 500	146 000	6 350	14 900	18 600	47 500	1 890	4 850	3 300	5 000

Note 1) Allowable minimum chamfer dimension r or r_1.
Remarks: This complex can support bi-directional axial load.

Bearing numbers	Abutment dimensions mm				Mass kg
	d_a min	D_a max	D_b max	r_{as} max	(approx.)
NKIA 5902	17.5	22	25.5	0.3	0.050
NKIA 5903	19.5	24	27.5	0.3	0.056
NKIA 5904	22.5	28	34.5	0.3	0.111
NKIA59/22	24.5	31	36.5	0.3	0.120
NKIA 5905	27.5	33	39.5	0.3	0.130
NKIA 5906	32.5	38	44.5	0.3	0.147
NKIA 5907	40	45	50	0.6	0.243
NKIA 5908	45	51	57	0.6	0.347
NKIA 5909	50	58	63	0.6	0.401
NKIA 5910	55	63	67	0.6	0.410
NKIA 5911	61	66.5	74	1	0.590
NKIA 5912	66	71.5	79	1	0.632
NKIA 5913	71	78.5	84	1	0.708
NKIA 5914	76	84	94	1	1.05

Type NKIB 59





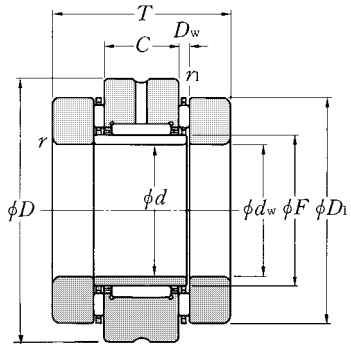
d 15～70mm

Boundary dimensions mm						Basic load ratings dynamic N	static N	dynamic kgf	static kgf	dynamic N	static N	dynamic kgf	static kgf	Limiting dimensions r/min	
						Radial				Axial				grease	oil
d	D	B $^{0}_{-0.3}$	C	F	$r_{s\,min}$ [1)]	C_r	C_{or}	C_r	C_{or}	C_a	C_{oa}	C_a	C_{oa}		
15	28	20	18	20	0.3	10 800	13 600	1 100	1 390	2 750	4 200	280	430	13 000	20 000
17	30	20	18	22	0.3	11 200	14 600	1 140	1 490	2 960	4 900	300	495	12 000	18 000
20	37	25	23	25	0.3	21 300	25 500	2 170	2 600	4 650	7 400	475	755	11 000	16 000
22	39	25	23	28	0.3	23 200	29 300	2 360	2 990	5 000	8 650	510	880	9 500	14 000
25	42	25	23	30	0.3	24 000	31 500	2 450	3 200	5 150	9 250	525	945	8 500	13 000
30	47	25	23	35	0.3	25 500	35 500	2 600	3 600	5 600	11 200	570	1 140	7 500	11 000
35	55	30	27	42	0.6	32 000	50 000	3 300	5 100	7 050	14 900	720	1 520	6 500	9 500
40	62	34	30	48	0.6	43 500	66 500	4 450	6 800	8 700	19 400	890	1 980	5 500	8 500
45	68	34	30	52	0.6	46 000	73 000	4 700	7 450	9 100	21 400	925	2 180	5 000	7 500
50	72	34	30	58	0.6	48 000	80 000	4 900	8 150	9 600	24 300	980	2 480	4 300	6 500
55	80	38	34	63	1	58 500	99 500	6 000	10 100	11 400	29 400	1 170	3 000	4 300	6 500
60	85	38	34	68	1	61 500	108 000	6 250	11 000	11 800	32 000	1 200	3 250	4 000	6 000
65	90	38	34	72	1	62 500	112 000	6 350	11 400	12 100	34 000	1 240	3 500	3 700	5 500
70	100	45	40	80	1	85 500	156 000	8 750	15 900	15 900	44 500	1 620	4 550	3 300	5 000

Note 1) Allowable minimum chamfer dimension r.
Remarks: 1. This complex can support bi-directional axial load.
2. The ball bearing of this complex bearing is complete with a synthetic resin cage, and the allowable operating temperature shall be 120˚C and, under continuous running, 100˚C or less.
3. This bearing type is the imported product from INA, Germany.

Bearing numbers	Abutment dimensions mm			Mass kg
	d_a min	D_a max	r_{as} max	(approx.)
NKIB 5902R	17.5	25.5	0.3	0.052
NKIB 5903R	19.5	27.5	0.3	0.058
NKIB 5904R	22.5	34.5	0.3	0.107
NKIB59/22R	24.5	36.5	0.3	0.122
NKIB 5905R	27.5	39.5	0.3	0.134
NKIB 5906R	32.5	44.5	0.3	0.151
NKIB 5907R	40	50	0.6	0.247
NKIB 5908R	45	57	0.6	0.320
NKIB 5909R	50	63	0.6	0.380
NKIB 5910R	55	67	0.6	0.385
NKIB 5911R	61	74	1	0.555
NKIB 5912R	66	79	1	0.595
NKIB 5913R	71	84	1	0.640
NKIB 5914R	76	94	1	0.985

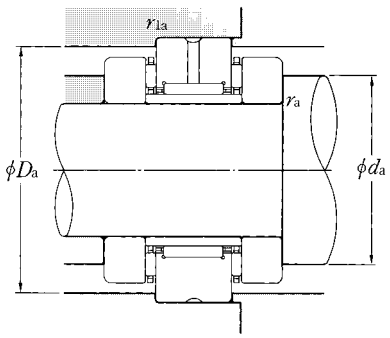
Type AXN



d 20～50mm

Boundary dimensions										Basic load ratings							
										dynamic	static	dynamic	static	dynamic	static	dynamic	static
										N		kgf		N		kgf	
										Radial				Axial			
d	d_w	D	D_1 $^{-0.20}_{-0.50}$	T $^{0}_{-0.370}$	C $^{0}_{-0.130}$	F	D_w	$r_{s\,min}$ [1]	$r_{1s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}	C_a	C_{oa}	C_a	C_{oa}
20	$20^{+0.061}_{+0.040}$	52	42	40	16	25	2	0.6	0.6	15 100	22 400	1 540	2 280	14 600	58 000	1 490	5 900
25	$25^{+0.061}_{+0.040}$	57	47	44	20	30	2	0.6	0.6	22 100	34 000	2 260	3 500	16 300	69 500	1 660	7 100
30	$30^{+0.061}_{+0.040}$	62	52	44	20	35	2	0.6	0.6	24 800	41 500	2 520	4 250	17 800	81 500	1 820	8 300
35	$35^{+0.075}_{+0.050}$	70	60	48	20	40	3	1	0.6	26 400	47 000	2 700	4 800	27 400	110 000	2 790	11 300
40	$40^{+0.075}_{+0.050}$	75	65	48	20	45	3	1	0.6	28 000	52 500	2 860	5 400	29 800	128 000	3 050	13 100
45	$45^{+0.075}_{+0.050}$	80	70	54	25	50	3	1	0.6	38 500	74 500	3 950	7 550	31 500	143 000	3 250	14 500
50	$50^{+0.075}_{+0.050}$	90	78	54	25	55	3	1	0.6	41 000	82 000	4 150	8 400	38 000	186 000	3 850	19 000

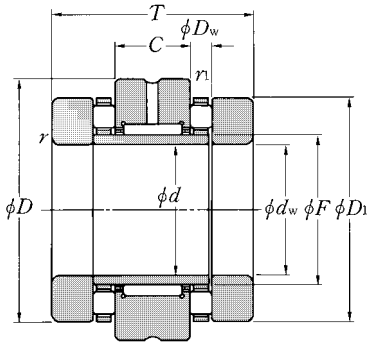
Note 1) Allowable minimum chamfer dimension r or r_1.



Limiting dimensions r/min grease	oil	Bearing numbers	Radial clearance μm min	max	Abutment dimensions mm d_a min	D_a max	r_{as} max	r_{1as} max	Preload N	kgf	Starting torque[2] N·mm	kgf·mm	Mass kg (approx.)
1 800	7 000	**AXN2052**	10	30	39	46	0.6	0.6	1 300	130	330	33	0.400
1 500	6 000	**AXN2557**	10	30	44	51	0.6	0.6	1 450	145	400	40	0.520
1 400	5 500	**AXN3062**	10	40	50	56	0.6	0.6	1 600	160	550	55	0.590
1 200	4 700	**AXN3570**	10	40	56	64	1	0.6	2 450	245	900	90	0.800
1 100	4 300	**AXN4075**	10	40	62	69	1	0.6	2 650	265	1 050	105	0.890
1 000	3 900	**AXN4580**	10	40	67	74	1	0.6	2 800	280	1 200	120	1.00
900	3 500	**AXN5090**	15	50	75	83	1	0.6	3 400	340	1 600	160	1.42

Note 2) Starting torque subject to standard preload.

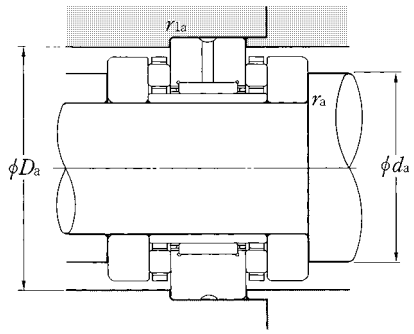
Type ARN



d 20～70mm

Boundary dimensions mm										Basic load ratings dynamic N Radial	static N Radial	dynamic kgf Radial	static kgf Radial	dynamic N Axial	static N Axial	dynamic kgf Axial	static kgf Axial
d	d_w	D	D_1 $^{-0.20}_{-0.50}$	T $^{0}_{-0.370}$	C $^{0}_{-0.130}$	F	D_w	$r_{s\,min}$ [1]	$r_{1s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}	C_a	C_{oa}	C_a	C_{oa}
20	$20^{+0.061}_{+0.040}$	52	42	46	16	25	5	0.6	0.6	15 100	22 400	1 540	2 280	27 300	68 000	2 780	6 900
20	$20^{+0.061}_{+0.040}$	62	52	60	20	30	7.5	1	0.6	22 100	34 000	2 260	3 500	53 500	129 000	5 450	13 100
25	$25^{+0.061}_{+0.040}$	57	47	50	20	30	5	0.6	0.6	22 100	34 000	2 260	3 500	27 800	72 500	2 840	7 400
25	$25^{+0.061}_{+0.040}$	72	62	60	20	35	7.5	1	0.6	24 800	41 500	2 520	4 250	54 500	139 000	5 550	14 200
30	$30^{+0.061}_{+0.040}$	62	52	50	20	35	5	0.6	0.6	24 800	41 500	2 520	4 250	31 000	87 000	3 150	8 900
30	$30^{+0.061}_{+0.040}$	80	68	66	20	40	9	1	0.6	26 400	47 000	2 700	4 800	74 500	190 000	7 600	19 400
35	$35^{+0.075}_{+0.050}$	70	60	54	20	40	6	1	0.6	26 400	47 000	2 700	4 800	43 000	121 000	4 350	12 400
35	$35^{+0.075}_{+0.050}$	85	73	66	20	45	9	1	0.6	28 000	52 500	2 860	5 400	82 000	222 000	8 350	22 600
40	$40^{+0.075}_{+0.050}$	75	65	54	20	45	6	1	0.6	28 000	52 500	2 860	5 400	45 500	135 000	4 650	13 800
40	$40^{+0.075}_{+0.050}$	90	78	75	25	50	9	1	0.6	38 500	74 500	3 950	7 550	85 000	238 000	8 650	24 200
45	$45^{+0.075}_{+0.050}$	80	70	60	25	50	6	1	0.6	38 500	74 500	3 950	7 550	48 000	150 000	4 900	15 300
45	$45^{+0.075}_{+0.050}$	105	90	82	25	55	11	1	0.6	41 000	82 000	4 150	8 400	121 000	340 000	12 300	34 500
50	$50^{+0.075}_{+0.050}$	90	78	60	25	55	6	1	0.6	41 000	82 000	4 150	8 400	62 500	215 000	6 350	21 900
50	$50^{+0.075}_{+0.050}$	110	95	82	25	60	11	1.1	0.6	41 000	85 000	4 200	8 700	125 000	365 000	12 800	37 000
55	$55^{+0.090}_{+0.060}$	115	100	82	25	65	11	1.1	0.6	44 500	98 000	4 550	10 000	130 000	385 000	13 200	39 500
60	$60^{+0.090}_{+0.060}$	120	105	82	25	70	11	1.1	0.6	45 000	91 500	4 600	9 350	134 000	410 000	13 700	42 000
65	$65^{+0.090}_{+0.060}$	125	110	82	25	75	11	1.1	0.6	55 000	104 000	5 600	10 600	138 000	435 000	14 100	44 500
70	$70^{+0.090}_{+0.060}$	130	115	82	25	80	11	1.1	0.6	57 000	119 000	5 800	12 200	142 000	460 000	14 500	47 000

Note 1) Allowable minimum chamfer dimension r or r_1.



Limiting dimensions r/min grease	oil	Bearing numbers	Radial clearance μm min	max	Abutment dimensions mm d_a min	D_a max	r_{as} max	r_{1as} max	Preload N	kgf	Starting torque[2] N·mm	kgf·mm	Mass kg (approx.)
1 800	7 000	**ARN2052T2**	10	30	39	46	0.6	0.6	2 500	250	430	43	0.440
1 500	6 000	**ARN2062**	10	30	48	56	1	0.6	4 950	495	1 150	115	0.910
1 500	6 000	**ARN2557T2**	10	30	44	51	0.6	0.6	2 600	260	500	50	0.560
1 200	4 900	**ARN2572**	10	40	56	66	1	0.6	5 050	505	1 400	140	1.22
1 400	5 500	**ARN3062T2**	10	40	49	56	0.6	0.6	2 900	290	650	65	0.630
1 100	4 400	**ARN3080**	10	40	63	73	1	0.6	6 900	690	2 100	210	1.54
1 200	4 800	**ARN3570T2**	10	40	56	64	1	0.6	3 950	395	1 050	105	0.850
1 000	4 100	**ARN3585**	10	40	68	77	1	0.6	7 600	760	2 500	250	1.67
1 100	4 400	**ARN4075T2**	10	40	61	69	1	0.6	4 200	420	1 250	125	0.930
950	3 800	**ARN4090**	10	40	73	87	1	0.6	7 850	785	2 850	285	2.15
1 000	4 000	**ARN4580T2**	10	40	66	74	1	0.6	4 450	445	1 550	155	1.16
850	3 300	**ARN45105**	15	50	83	96	1	0.6	11 200	1 120	4 350	435	3.16
900	3 600	**ARN5090**	15	50	75	83	1	0.6	5 800	580	2 050	205	1.48
800	3 100	**ARN50110**	15	50	88	101	1	0.6	11 600	1 160	4 900	490	3.38
750	2 900	**ARN55115**	15	50	93	106	1	0.6	12 000	1 200	5 500	550	3.61
700	2 700	**ARN60120**	15	50	98	111	1	0.6	12 400	1 240	6 000	600	3.81
650	2 600	**ARN65125**	15	50	103	116	1	0.6	12 800	1 280	6 500	650	4.00
650	2 500	**ARN70130**	15	50	106	121	1	0.6	13 200	1 320	7 000	700	4.25

Note 2) Starting torque subject to standard preload.

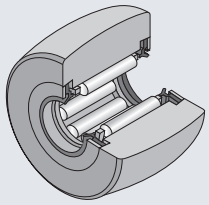
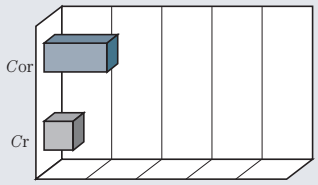
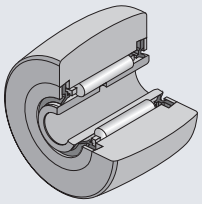
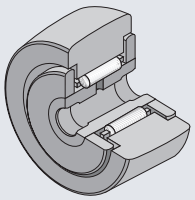
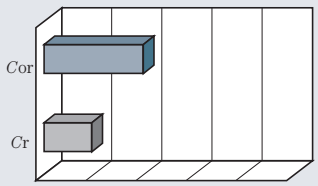
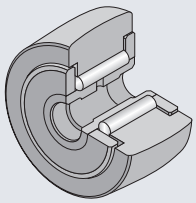
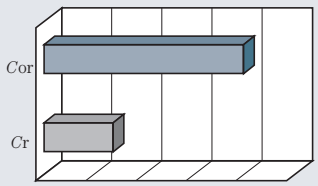
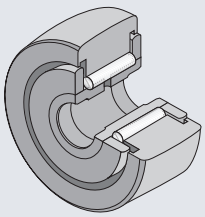
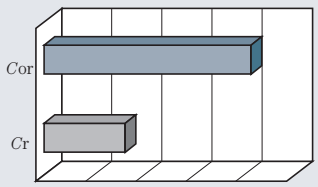
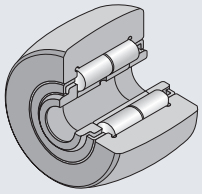
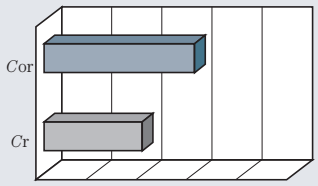
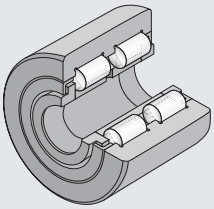
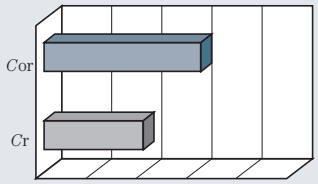
Roller Followers / Cam Followers



Roller Followers (Yoke Type Track Rollers)

NTN Yoke Type Track Rollers is the rolling mechanisms whose outer ring rolls on a track. For example, these track rollers are applied to eccentric roller, guide roller, rocker arm roller, cam roller and pressure roller. For that, the outer ring is designed to such a wall thickness as to be resistible to high load and shock load.

Both of spherical outer surface (rolling surface) and cylindrical outer surface are available for the outer ring. The spherical outer ring can damp edge-load acting on the contact surface between the track and the track roller, while the cylindrical outer ring (Tail code: **X**) has track load capacity greater than the spherical outer ring.

Bearing type	Applied shaft diameter (mm)	Load capacity	Composition of nominal bearing number
RNA22 	ϕ6～ϕ58		**RNA 22 / 6 LL** Tail code LL: Seal Dimension code Dimension series code Type code
NA22 			**NA 22 06 X LL** Tail code LL: Seal Tail code X: Cylinder outer diameter Dimension code Dimension series code Type code
NATR 	ϕ5～ϕ50		**NATR 30 X LL** Tail code LL: Seal Tail code X: Cylinder outer diameter Dimension code Type code
NATV 	ϕ5～ϕ50		**NATV 25 LL** Tail code LL: Seal Dimension code Type code
NACV 	ϕ6.35～ϕ57.15		**NACV 32 X LL** Tail code LL: Seal Tail code X: Cylinder outer diameter Dimension code Type code
NUTR 	ϕ15～ϕ50		**NUTR 3 10** Dimension code Dimension series code Type code
NUTW 	ϕ15～ϕ50		**NUTW 2 05 X** Tail code X: Cylinder outer diameter Dimension code Dimension series code Type code

Each listed load capacity is subject to reference bearing bore diameter of ϕ50 or ϕ50.8

Track roller components	Features
Inscribed circle diameter : ϕ6 Type with cage Inner ring : without Outer profile : spherical Seal : with	● The needle rollers and the cage are retained in the outer ring with a steel plate reinforced synthetic rubber seal, and these are inseparable from each other. ● The needle rollers guided by cage. ● Shaft (Pin) must be provided with a thrust washer or a flange to guide the outer ring.
Inscribed circle diameter : ϕ30 Type with cage Inner ring : with Outer profile : cylindrical Seal : with	
Inscribed circle diameter : ϕ30 Type with cage Outer profile : cylindrical Seal : with	● Needle rollers guided by cage. ● The outer ring is guided in axial direction by the thrust washer press-fitted in the inner ring. ● Labyrinth is formed between the outer ring and the thrust washer.
Inscribed circle diameter : ϕ30 Full complement roller type Outer profile : cylindrical Seal : with	● The outer ring is guided in axial direction by the thrust washer press-fitted in the inner ring. ● Great rated load due to the full complement roller type. ● Lower allowable running speed than bearing with cage. ● Labyrinth is formed between the outer ring and the thrust washer.
Inscribed circle diameter : ϕ15.875 Full complement roller type Outer profile : cylindrical Seal : with	
Inscribed circle diameter : ϕ50 Double-row cylindrical Full-complement roller type Labyrinth seal Outer profile : spherical	● Greater rated load, best-suited to applications subjected to high load and shock load. ● A steel plate is press-fitted in the outer ring to form labyrinth between the plate and the side plates at both sides of the inner ring, whereby the required sealing performance is secured and the side plates are held so as not to separate from one another. ● The inner ring and the side plates are tightened together in axial direction for non-clearance. ● The outer ring is guided in axial direction by the outer ring ribs and the end faces of cylindrical roller. Particularly Type NUTW provides the following features. ● The greatest rated load, of the roller followers. ● Due to the outer ring with inner ribs, this type is excellently resistible to axial load and moment load and, in addition, runs smoothly, though depending on actual operatingconditions. ● Good lubrication effect and longer life are expectable by increasing thegrease fill volume.
Inscribed circle diameter : ϕ25 Double-row cylindrical Full-complement roller type with center rib Labyrinth seal Outer profile : spherical	